

Resume — Tims Pečerskis, MSc

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Systems engineer, algorithm designer, and product builder. 15+ years designing novel algorithms under real-time, resource-constrained conditions and shipping them as production systems — **from bare metal signal processing through distributed GPU orchestration**. Proven ability to lead cross-discipline teams, manage multi-client product portfolios, own end-to-end product lifecycles from user research through delivery, and drive technical sales from presales through production support. Published researcher, patent author, and conference speaker with a track record of consistently outperforming project timelines.

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Domain Expertise

Product & Project Leadership — Product design · product strategy · user research · roadmap ownership · reliability & scalability planning · SDLC and hardware lifecycle management · agile · compliance & certification · cross-functional team leadership · hiring & org scaling · performance engineering · reliability engineering · portfolio management · multi-client delivery.

Algorithm Design & Systems Engineering — Real-time signal processing · low-latency systems · deterministic processing · hardware-software co-design · latency profiling · machine learning · protocol security · software engineering · electrodynamics · data analysis · R&D · patent development and IP strategy.

Business Operations — Budget ownership · process optimisation · vendor management · stakeholder communication · compliance framework navigation.

Growth — Technical sales · presales · account management · solutions architecture · developer advocacy · developer experience design · product-led growth · public speaking · technical writing.

Professional Experience

Founder — [Peeramid Labs](#)

November 2024 — Present

Lean team of three, experimenting across AI infrastructure, blockchain security, and privacy.

Skills: Rust · C · CUDA · Python (pandas, scipy, sklearn) · Huggingface · vLLM · SGLang · NATS.io · Docker · Redis · Postgres · GCP · TS/JS (React, Node) · LaTeX · Solidity · EVM (Arbitrum, Ethereum, Reth, Celo, Optimism, zkSync, Base, smart accounts, MEV) · Circom · Foundry · wagmi · alloy-rs · Zama FHE · Web3 security · protocol engineering · game-theoretic and financial modelling · P2P networking · business development · investor relations · product strategy for compliance and regulation.

Achievements:

- Shipped three ML products end-to-end — defining product scope, designing algorithms, and leading implementation: [AI code-slop firewall](#), smart-contract audit tool (**83% precision**), and a health-prediction engine ($R^2 = 0.52$ on held-out subjects).
- Designed **Noolog** end-to-end as a product — from research paper to shipped platform. Sovereign agent-to-agent reasoning network implementing NSED: agents propose, cross-evaluate, and converge with full audit trace. Event-driven low-latency Rust system on NATS JetStream — immutable per-round traces, live SSE streaming, runtime human oversight and kill switch. Led a team of three across architecture, compliance, and product design. **84% on AIME 2025** with three 8–20B local models, matching DeepSeek-R1 on **4x less VRAM**.
- Authored [Mixture-of-Models: Unifying Heterogeneous Agents via N-Way Self-Evaluating Deliberation](#) (NSED, arXiv:2601.16863) — a distributed GPU neural architecture for heterogeneous model orchestration delivering **4x CapEx reduction** or **30% accuracy boost** on any base model benchmarked.
- Built the production Rust implementation of NSED on NATS JetStream. Open-sourced the A2A agent SDK [quorum-rs](#) on crates.io with docs served to humans and agents over MCP at [noolog.io](#).

- Designed the Noolog product experience: Rust-native desktop client, compliance-native architecture generating **EU AI Act / GDPR / SOC2** audit artifacts from deliberation traces, developer SDK, and MCP integration for agent-facing documentation.
- Filed **6 patents across 5 families** — multi-agent AI coordination, novel AI architecture (zero prior art), entropy-based model reputation systems, cryptographic compliance infrastructure, and hardware-bound licensing.
- Identified and responsibly disclosed a critical vulnerability in **Ethereum EIP-7702** — [full write-up](#).

Lecturer, Blockchain Architecture & Programming — [Invisible Garden Foundation](#), US

October 2024 — Present

Skills: Tutoring · EVM · Solidity · computer science · business strategy advisory · startup incubation · zero-knowledge proofs · FHE · Zama.

Achievements:

- Tutored **100+ students** over two years in blockchain fundamentals, smart-contract security, and Solidity. Curriculum design and delivery covering EVM, zero-knowledge proofs, and FHE.

Solution Architect / Senior Solution Developer — [OpenZeppelin](#), UK

November 2022 — August 2024

Skills: AWS · KMS · Terraform · DataDog · Grafana · IaC · technical account management · pre-sales · solution engineering · solutions architecture · public speaking · incident response planning · audits · DevSecOps · security monitoring · governance monitoring · DeFi · Cosmos SDK · Solidity · Rust · wargames.

Achievements:

- Led **technical account management and pre-sales** for the customer portfolio — owning discovery, scoping, product-fit qualification, and post-sale delivery handoff. Translated customer security requirements into deployable product configurations on the Defender platform.
- Served as technical advisor on the BIS [Project Mariana](#), coordinating across institutional stakeholders to contribute to the successful launch of the world's largest cross-institutional banking CBDC prototype.
- Co-authored the [EEA DeFi Risk Assessment Guidelines](#) — an industry-standard technical specification shaping how protocols assess and communicate risk.
- Led deployment of real-time monitoring and automated incident-response infrastructure on OpenZeppelin's IaC [Defender](#) platform, delivering near-100% continuous risk-factor monitoring coverage for customers.
- **Zero hacks on my customer portfolio during my shift.**

Product Engineer — [Moonstream](#) (formerly *Simiotics / Bugout*), California, US

June 2021 — August 2022

Skills: JS · TS · React · web3.js · ethers.js · Brownie · Figma · product design · UX research · big-data analysis.

Achievements:

- Shipped two products from zero in a fast-paced startup environment:
 - **Bugout** — developer-tooling analytics platform.
 - [Moonstream](#) — product design and UX research for drop-in game economy mechanics (lootboxes, crafting, achievements, leaderboards) aimed at quick developer integration. Platform processed **\$4B+ in transaction volume** in its first months post-launch.

Head of Engineering — SIA TeknoLogic, Latvia

February 2016 — November 2020

Owned the OEM/ODM product portfolio for all key accounts. Led cross-functional teams spanning hardware, firmware, certification, supply chain, and business development. Delivered **30+ distinct hardware/software products** end-to-end (concept → production) with near-100% customer retention.

Skills: C · C++ · Assembly · embedded Linux · Yocto · RTOS · Qt · Altium Designer · KiCad · high-

speed digital / power / mixed / analog / PCI-E system design · schematic and PCB · BoM management · ARM-M and ARM-A · Nordic nRF · STM32 · IAR · GCC · CMake · EMI/EMC certification · DFM · supply-chain development · sales · business development · resource planning.

Achievements:

- **Erica Synths** — co-defined the product specification for the **Zen Delay** with Ninja Tune, translating artist requirements into technical constraints. Led cross-functional delivery across hardware, firmware, and manufacturing. Designed the DSP module — hardware and algorithm — with clock-level optimisations achieving a **30% BoM cost reduction** against competitors. Top reviews from [Coldcut](#), The Bug, Toro y Moi and others.
- **Erica Synths** — led mixed-signal system design for the **Black Dual VCF** around the Curtis CEM3320 filter chip re-issued as the **AS3320** — bridging digital control algorithms and analog signal path.
- **Erica Synths** — designed the **DIY Delay full kit** as a platform product anchoring Erica Synths' developer and hacker community — product strategy driving community growth.
- **Aretai Speakers** — owned the hardware product portfolio; **Aretai Contra received the Red Dot Award in 2022**. Hardware included precision electromechanical systems requiring tight tolerancing.
- **Coastal Amusements** — led the full product redesign of the amusement-machine motherboard, managing cross-functional delivery across schematic, PCB, firmware, and certification. Designed real-time multi-bus control system — **>200A load**, automotive-grade security over I²C, SPI and USB. Engineered as a modular product platform supporting arbitrary customer business objectives.
- **LeClanché** — delivered the **Battery Management System** power circuitry for high-reliability lithium battery packs.
- **Barix** — led development and full **EMI/EMC compliance certification** for the [Barix Exstreamer M400](#) audio-over-IP appliance.

Product Owner — [Edge Autonomy](#) (acquired by [Redwire](#), NYSE: RDW)

March 2015 — February 2016

Product Owner managing a cross-discipline team of **15 engineers** spanning electronics and mechanical. Owned the product roadmap, defined priorities, and set up the company's documentation and delivery infrastructure.

Skills: Radio engineering · product ownership · documentation · training · leadership · project management · resource planning.

Achievements:

- Increased operational range of video-streaming radio links from **5 km to 140 km+** by fixing design flaws in the existing platform. System operated under strict latency and throughput constraints for live video relay.
- Cleared a **2-year product backlog in a single quarter** by ruthlessly triaging scope and aligning engineering priorities with product roadmap.
- Outperformed the project roadmap by **3x**.

Microwave / R&D Engineer — [SAF Tehnika](#)

June 2011 — February 2015

Core contributor to the company's flagship [Integra](#) microwave radio. Combined algorithm design with RF systems engineering to create new product categories and reduce manufacturing costs.

Skills: Microwave engineering (5.6–60 GHz) · low-noise amplifiers · passive circuits · microstrip design · CST Studio · Matlab · RF synthesis (Genesys) · Altium Designer · documentation · LoRa · DFM.

Achievements:

- Designed and patented the core signal-processing algorithm for the **Spectrum Compact** — a world-first hand-held battery-powered spectrum analyzer that became SAF Tehnika's best-earning product. Real-time processing under strict power and size constraints.
- Core contributor to the [Integra](#) microwave radio backbone: designed microwave filters, orthomode transducer (OMT), and led R&D validation for orbital angular momentum antenna concepts.
- Designed microwave circuits for flagship products — ultra-wide-bandwidth **3–6 GHz amplifiers and LNAs**, mixers, polarisation splitters, microwave filters, waveguides, antennas. Typically outperformed timelines by **~2x**.

- Applied Monte-Carlo statistical modeling to manufacturing tolerance analysis, optimizing yield under tight RF performance constraints. Achieved **30%+ cost reduction**.

Patents & Publications

- **Mixture-of-Models: Unifying Heterogeneous Agents via N-Way Self-Evaluating Deliberation.** 2026. doi.org/10.5281/zenodo.18338777
- **EEA DeFi Risk Assessment Guidelines — Version 1.** EEA Technical Specification, 17 July 2024.
- **Spectrum Analyzer of Ultra-Low Intermediate Frequency.** LV 14855A, 2014, SAF Tehnika.
- **6 patents filed across 5 families** — multi-agent AI coordination · novel AI architecture (zero prior art) · entropy-based model reputation · cryptographic compliance infrastructure · hardware-bound licensing.

Education

- **MSc, Electrical and Electronics Engineering** — [Riga Technical University](#).
- **BSc, Electrical and Electronics Engineering** — [Riga Technical University](#).

Languages

- English — Full Professional
- Latvian — Bilingual
- Russian — Bilingual
- Spanish — Limited
- Chinese — Elementary (studying)

After Work

Watersports (surfing, diving, freediving) · cycling · yoga · hiking · climbing · music (production, [SoundCloud](#)) · woodworking · retro / MMORPG gaming (retired WoW / L2 raid master).